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Experiment 8: Linear Regression (LR)

Aim

To implement the **Linear Regression algorithm** in Python and predict continuous values.

Python Program

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load California Housing Dataset
housing = fetch_california_housing()

X = housing.data
y = housing.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

print("First 10 Actual Values:")
```

```
print(y_test[:10])

print("\nFirst 10 Predicted Values:")
print(y_pred[:10])

print("\nMean Squared Error:")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))

# Predict for a new sample
new_sample = [X_test[0]]
prediction = model.predict(new_sample)

print("\nPredicted House Price:")
print(prediction[0])
```

Sample Output

```
... First 10 Actual Values:
    [0.477   0.458   5.00001 2.186   2.78    1.587   1.982   1.575   3.4
     4.466  ]

    First 10 Predicted Values:
    [0.72604907 1.76743383 2.71092161 2.83514727 2.60695807 2.01073856
     2.64067386 2.16706161 2.74012056 3.90361526]

    Mean Squared Error:
    0.5305677824766757

    R2 Score:
    0.595770232606166

    Predicted House Price:
    0.7260490726243773
```